

TOPIC 1: ECOSYSTEM'S CARRYING CAPACITY AND POPULATION GROWTH

1. What term describes the largest number of individuals an ecosystem can support sustainably over time without damaging its resources?

Answer: Carrying capacity

2. What do we call any factor that limits or stops a population from growing further?

Answer: Limiting factor

3. Which of these is not a typical limiting factor for animal populations: food, water, shelter, or body color?

Answer: Body color of individuals

4. What name is given to rapid, unrestricted population growth that produces a steep upward curve?

Answer: Exponential growth / J-shaped curve

5. What happens to a population's growth rate as it gets close to the carrying capacity of its environment?

Answer: It slows down and approaches zero

6. What is the term for the growth pattern that rises quickly then levels off smoothly at carrying capacity?

Answer: Logistic growth

7. What effect does a shortage of available mates have on the size of a small population?

Answer: It limits population growth

8. What does the flat, horizontal section at the top of a logistic growth curve stand for?

Answer: Carrying capacity / stable population size

9. Drought reduces food and water, causing rabbit numbers to drop sharply. What kind of factor is drought in this case?

Answer: Limiting factor

10. What phrase refers to the count of individuals living within a specific unit of area or volume?

Answer: Population density

11. A forest normally supports 500 deer. After a typhoon destroys many trees, what will happen to its carrying capacity?

Answer: It will decrease

12. What type of limiting factor has a stronger effect when a population is large and crowded?

Answer: Density-dependent limiting factor

13. What type of limiting factor affects populations regardless of whether they are dense or sparse?

Answer: Density-independent limiting factor

14. What is the main reason population growth slows as it nears carrying capacity?

Answer: Competition for resources becomes more intense

15. Rats increase in number, so snakes that eat them also rise, slowing rat population growth. Predation acts as what kind of limiting factor?

Answer: Biotic limiting factor

16. In nature, what eventually happens to a population showing J-shaped exponential growth?

Answer: Limiting factors stop growth; it either crashes or levels into an S-shaped curve

17. If a population is smaller than the environment's carrying capacity, how will its size usually change?

Answer: It will increase until it approaches carrying capacity

18. Building a landfill destroys most of a frog's breeding pond. Which limiting factor is most reduced?

Answer: Shelter / breeding space

19. Which pair consists only of abiotic (non-living) limiting factors?

Answer: Water availability and temperature extremes

20. Which growth pattern best matches the current human population?

Answer: Rising J-shaped curve / early logistic growth

21. Arrange the stages of logistic growth in the correct order.

Answer: Slow initial growth → rapid exponential growth → slowing growth → stable at carrying capacity

22. What usually happens after a population temporarily exceeds its carrying capacity?

Answer: Population crash or sharp decline due to resource shortage

23. A lake can support 1,000 tilapia. If 1,200 are added, what will most likely happen?

Answer: Many will die; numbers will settle back near 1,000

24. Why is the availability of mates especially important for small, isolated groups?

Answer: It reduces the chance of successful reproduction

25. What is the key difference between density-dependent and density-independent limiting factors?

Answer: Density-dependent impact rises with crowding; density-independent impact is the same at any population size

26. Give an example that shows growth slowing as it nears carrying capacity.

Answer: Mouse numbers change from 100 → 180 → 210 → 215 → 212

27. How does competition limit population size near carrying capacity?

Answer: Fewer individuals survive and reproduce because resources become scarce

28. Protecting both food sources and nesting sites will raise the carrying capacity of the Philippine Eagle. What does this show about carrying capacity?

Answer: It depends on multiple limiting factors working together

29. Why can exponential growth never continue indefinitely in any real ecosystem?

Answer: All natural resources are finite

30. Under what condition does a population remain stable at carrying capacity?

Answer: When birth rate is roughly equal to death rate

TOPIC 2: BIOTECHNOLOGY

1. What is the general term for using living things or biological processes to create useful goods?

Answer: Biotechnology

2. What process forms the basis of traditional biotechnology used to make foods like cheese and vinegar?

Answer: Fermentation

3. What Filipino food product is made when bacteria ferment coconut water?

Answer: Nata de coco

4. What acid is produced when alcohol is further oxidized to form vinegar?

Answer: Acetic acid

5. What acid is produced by bacteria to make yogurt, cheese, and pickles?

Answer: Lactic acid

6. What main ingredients are fermented to produce soy sauce?

Answer: Soybeans and usually wheat or grain

7. What part of an organism is deliberately altered to create a Genetically Modified Organism?

Answer: DNA / genetic material

8. What does "in vitro" mean in the process of In Vitro Fertilization?

Answer: Fertilization happens outside the body in a laboratory

9. What is one common benefit of genetically modified crops?

Answer: Pest resistance / higher yield / reduced pesticide use

10. What is one major ethical concern linked to IVF?

Answer: Moral status of embryos / cost and fair access

11. Which list contains only products made using traditional fermentation?

Answer: Cheese, soy sauce, vinegar, nata de coco, yogurt

12. What gas does yeast produce that makes bread rise?

Answer: Carbon dioxide

13. Golden Rice is modified to produce high levels of what nutrient to prevent blindness?

Answer: Beta-carotene (Vitamin A precursor)

14. What is one possible environmental risk of planting insect-resistant GMO crops?

Answer: Pests may develop resistance / harm to beneficial insects

15. What is one economic concern regarding GMO seeds in the Philippines?

Answer: High cost and corporate control affecting small farmers

16. How is human insulin for diabetes produced using modern biotechnology?

Answer: The human insulin gene is inserted into bacteria that make the insulin

17. Which microbe is responsible for turning alcohol into acetic acid?

Answer: Acetic acid bacteria

18. Why is fermentation considered an ancient form of biotechnology?

Answer: It was used for thousands of years before microbes were discovered

19. What is true about the genetic makeup of an organism produced by cloning?

Answer: It is genetically identical to the donor parent

20. What is one major ethical concern about using gene-editing tools like CRISPR on humans?

Answer: Risk of unintended genetic changes / creation of "designer babies"

21. What is the correct two-step process for making traditional vinegar?

Answer: Sugar → (yeast) → ethanol → (bacteria) → acetic acid

22. What material forms the solid jelly part of nata de coco?

Answer: Bacterial cellulose

23. What international agreement governs the safe handling and movement of living GMOs?

Answer: Cartagena Protocol

24. What is one environmental benefit of some approved GMO crops?

Answer: Less insecticide use and reduced chemical pollution

25. What issue involves both ethics and economics regarding IVF in the Philippines?

Answer: High cost makes it unavailable to many people

26. How does fermentation help preserve food?

Answer: It produces acids or alcohol that stop harmful microbes from growing

27. Where does the gene that gives Bt corn its pest-killing ability come from?

Answer: The bacterium *Bacillus thuringiensis*

28. What is the main difference between traditional and modern biotechnology?

Answer: Traditional uses natural processes; modern directly modifies DNA in a lab

29. What ethical concern arises because several embryos are created during IVF?

Answer: What happens to unused or extra embryos

30. Give one example of a practice that can be done using both traditional and modern biotechnology.

Answer: Plant breeding

TOPIC 3: PLATE TECTONICS

1. What theory states that Earth's outer layer is made of large moving slabs of rock?

Answer: Plate Tectonics

2. What name is given to these large moving slabs?

Answer: Tectonic / lithospheric plates

3. Approximately how fast do tectonic plates move each year?

Answer: About 2–10 centimeters per year (as fast as fingernails grow)

4. What modern technology is used to measure plate movement accurately?

Answer: GPS satellites

5. What kind of boundary forms where plates move apart?

Answer: Divergent boundary

6. What kind of boundary forms where plates collide?

Answer: Convergent boundary

7. What kind of boundary forms where plates slide past each other?

Answer: Transform fault boundary

8. What layer of the mantle allows plates to move over it?

Answer: Asthenosphere

9. What process occurs when one plate sinks beneath another into the mantle?

Answer: Subduction

10. What is the name of the major fault system running through the Philippines?

Answer: Philippine Fault System

11. What landform is created when two continents collide?

Answer: Large non-volcanic mountain ranges

12. What features form when an oceanic plate subducts under a continental plate?

Answer: Deep ocean trench and volcanic mountains on land

13. Between which two major plates does the Philippines lie?

Answer: Philippine Sea Plate and Eurasian Plate

14. Why does the Philippines have many volcanoes and trenches?

Answer: Multiple plates are subducting around the archipelago

15. What geologic event is most common at transform boundaries?

Answer: Earthquakes

16. How were the Himalayan Mountains formed?

Answer: Collision of the Indian Plate with the Eurasian Plate

17. How did the Andes Mountains form?

Answer: Subduction of the Nazca Plate under the South American Plate

18. What is the main force that drives plate movement?

Answer: Thermal convection currents in the mantle

19. What happens to the overriding plate when another plate subducts beneath it?

Answer: Magma rises to form volcanoes; folding and faulting create mountains

20. Where are new sections of ocean floor created?

Answer: At divergent boundaries / mid-ocean ridges

21. If the Philippine Sea Plate keeps moving northwest, where will the Philippines be in 50 million years?

Answer: About 3,500 km northwest, joining the eastern edge of Asia

22. Why is GPS preferred over older methods for measuring plate motion?

Answer: It gives highly precise, continuous, global measurements

23. What process formed the Mariana Trench and Philippine Trench?

Answer: Subduction

24. What type of boundary created Mayon Volcano and the Manila Trench?

Answer: Convergent boundary

25. Why do no volcanoes form when two continental plates collide?

Answer: Continental crust is too light to sink and melt

26. What property allows the asthenosphere to flow slowly and move the plates?

Answer: It is solid rock that is soft and flexible over very long periods

27. What does having trenches on both sides of the Philippines tell us about its tectonic setting?

Answer: It is squeezed between two sets of subducting plates

28. What kind of fault is the Philippine Fault System?

Answer: Left-lateral transform fault

29. What is the difference between landforms formed by ocean-ocean and ocean-continent subduction?

Answer: Ocean-ocean forms volcanic island arcs; ocean-continent forms volcanic mountains on land

30. If we reversed plate motion 50 million years ago, where would India and the Philippines be?

Answer: India was an island south of Asia; the Philippines was separate volcanic islands

TOPIC 4: GLOBAL CLIMATE

1. What term describes the average weather pattern of an area over many years?

Answer: Climate

2. What is the gradual rise in Earth's average surface temperature called?

Answer: Global warming

3. What broader term includes warming plus changes in rainfall, storms, and seasons?

Answer: Climate change

4. What name is given to gases that trap heat in the atmosphere?

Answer: Greenhouse gases

5. What is the main human-caused greenhouse gas driving modern warming?

Answer: Carbon dioxide (CO₂)

6. Name two natural greenhouse gases that keep Earth warm enough for life.

Answer: Water vapor and carbon dioxide

7. What does the "enhanced greenhouse effect" mean?

Answer: Extra greenhouse gases from human activities trap more heat than normal

8. What direct effect do melting glaciers and ice sheets have on oceans?

Answer: Sea level rise

9. What do ice core samples show about carbon dioxide levels today compared to the past 800,000 years?

Answer: They are far higher than any previous natural level

10. Aside from melting ice, what other process causes sea levels to rise?

Answer: Thermal expansion of seawater as it warms

11. Which list contains only major greenhouse gases?

Answer: Carbon dioxide, methane, nitrous oxide, CFCs, water vapor

12. What human activities produce large amounts of methane?

Answer: Livestock, rice farming, landfills, leaking natural gas

13. What two harmful effects do CFCs cause?

Answer: They trap heat and destroy the ozone layer

14. Give three long-term signs that climate change is happening.

Answer: Rising temperatures, shrinking ice, rising sea levels

15. How does cutting down forests worsen global warming?

Answer: It releases stored carbon and removes trees that absorb CO₂

16. Is a single hot day or one storm enough proof of climate change?

Answer: No; climate change requires long-term trends over many years

17. What is the correct sequence of events leading to human-caused warming?

Answer: Burning fossil fuels → extra CO₂ → stronger greenhouse effect → higher temperatures

18. Why will warming continue for decades even if all emissions stop now?

Answer: Greenhouse gases stay in the air for a long time and oceans hold heat

19. What human activity produces the most greenhouse gases?

Answer: Burning fossil fuels for energy and transport

20. How do scientists confirm that extra CO₂ comes from burning fossil fuels?

Answer: Its chemical makeup matches carbon from coal, oil, and gas

21. What is the difference between the natural and enhanced greenhouse effect?

Answer: Natural keeps Earth livable; enhanced is caused by humans and warms the planet too much

22. What kind of feedback loop happens when melting sea ice leads to more heat being absorbed?

Answer: Positive feedback

23. Why is a 1°C rise in global average temperature a serious problem?

Answer: It causes much larger increases in extreme weather and uneven warming

24. How does global warming affect tropical cyclones in the Philippines?

Answer: They become stronger and carry more rainfall

25. Why does the Arctic warm faster than other parts of the world?

Answer: Ice-albedo positive feedback: less ice means more sunlight absorbed

26. Why can't natural factors explain the rapid warming seen recently?

Answer: Natural factors alone would cause cooling; only human greenhouse gases match the trend

27. How does extra CO₂ harm marine life?

Answer: It makes seawater more acidic, damaging coral and shellfish

28. Give three observed signs of climate change in the Philippines.

Answer: Rising sea levels, stronger typhoons, longer dry spells

29. Why is water vapor not considered the main cause of modern warming?

Answer: It increases because of warming; it does not start the warming

30. What relationship do ice core records show between CO₂ and global temperature?

Answer: They rise and fall together; current levels are far higher than natural peaks

TOPIC 5: GLOBAL INTERACTIONS

1. What does the abbreviation ENSO stand for?

Answer: El Niño Southern Oscillation

2. What is the warm phase of the ENSO cycle called?

Answer: El Niño

3. What is the cool phase of the ENSO cycle called?

Answer: La Niña

4. In which ocean basin does the ENSO cycle occur?

Answer: Tropical Pacific Ocean

5. What are the steady winds that normally blow east to west across the tropical Pacific?

Answer: Trade winds

6. What is the most common effect of El Niño on the Philippines?

Answer: Prolonged drought and below-normal rainfall

7. What is the most common effect of La Niña on the Philippines?

Answer: Above-normal rain, more floods, and more typhoons

8. What does the "Southern Oscillation" part of the name refer to?

Answer: Shifts in air pressure between the eastern and western Pacific

9. How long do El Niño and La Niña events typically last?

Answer: About 9–12 months, sometimes up to 2 years

10. Which agency officially declares El Niño or La Niña in the Philippines?

Answer: PAGASA

11. What happens to trade winds when an El Niño begins?

Answer: They weaken or reverse direction

12. What happens to the pool of warm ocean water during El Niño?

Answer: It shifts toward the central and eastern Pacific

13. Why does the Philippines usually get fewer typhoons during El Niño?

Answer: Storms form farther away and wind patterns suppress development

14. Why does the Philippines usually get more typhoons during La Niña?

Answer: Conditions become more favorable; storms form closer to the country

15. Which Philippine industries are most harmed by El Niño drought?

Answer: Farming: rice, corn, and coconut production

16. List three common impacts of El Niño in the Philippines.

Answer: Drought, water shortage, crop damage

17. List three common impacts of La Niña in the Philippines.

Answer: Heavier rain, more floods, more typhoons

18. Why do different countries experience different effects from El Niño and La Niña?

Answer: They change global wind and rain patterns differently in each place

19. Why do government agencies monitor ENSO closely?

Answer: To issue warnings and prepare for disasters

20. What does a "neutral" ENSO condition mean?

Answer: Ocean and weather conditions are near their long-term average

21. What is the Walker Circulation?

Answer: The large east-west cycle of rising and sinking air across the tropical Pacific

22. Why does rain fall on the eastern Pacific instead of the Philippines during El Niño?

Answer: The main rain belt follows the shifted warm water eastward

23. How does climate change affect El Niño and La Niña events?

Answer: It makes them stronger and more frequent

24. What is the correct sequence leading to dry weather in the Philippines during El Niño?

Answer: Trade winds weaken → warm water shifts east → rain belt moves away → less rainfall

25. How does El Niño affect Metro Manila's water supply?

Answer: It reduces rainfall that fills major dams like Angat and Ipo

26. How do El Niño and La Niña change typhoon paths toward the Philippines?

Answer: El Niño: storms turn away; La Niña: more storms hit land

27. What were some major effects of the strong 2015–2016 El Niño in the Philippines?

Answer: Severe drought, record heat, and large crop losses

28. How do El Niño and La Niña affect the monsoon seasons?

Answer: El Niño weakens the southwest monsoon; La Niña strengthens it

29. Why do impacts vary from one El Niño event to another?

Answer: Differences in strength, location of warm water, and background climate

30. Why is it important for students to learn about ENSO?

Answer: It helps understand and prepare for changes in food, water, and weather

TOPIC 6: GLOBAL AND LOCAL SUSTAINABILITY

1. What term describes actions that reduce or prevent greenhouse gas emissions?

Answer: Mitigation

2. What term describes actions that adjust to changes already happening?

Answer: Adaptation

3. What renewable energy uses heat from underground, abundant in the Philippines?

Answer: Geothermal energy

4. What energy source uses flowing or falling water?

Answer: Hydroelectric power / hydropower

5. What renewable energy comes directly from sunlight?

Answer: Solar energy

6. What is the biggest source of greenhouse gas emissions in the Philippines?

Answer: Burning fossil fuels like coal, oil, and gas

7. What simple action most reduces transport emissions?

Answer: Walk, bike, or take public transport instead of private cars

8. What do the three "R's" of waste management stand for?

Answer: Reduce, Reuse, Recycle

9. How do trees help fight climate change?

Answer: They absorb and store carbon dioxide

10. Give one observed effect of climate change already seen in the Philippines.

Answer: Stronger typhoons / longer droughts / rising sea levels

11. Which list contains renewable energy sources found in the Philippines?

Answer: Geothermal, hydro, solar, wind, biomass

12. How does switching to renewable energy help fight global warming?

Answer: It produces almost no carbon dioxide

13. What is one effect of rising sea levels on Philippine coastal communities?

Answer: Coastal flooding, erosion, and saltwater damage

14. Why does using energy-efficient appliances help the climate?

Answer: Less electricity used means less fuel burned at power plants

15. What is one effective way households can lower their carbon footprint?

Answer: Reduce wasted electricity and use efficient appliances

16. How does separating and composting waste help reduce greenhouse gases?

Answer: It prevents methane from forming in landfills

17. What is the purpose of the Renewable Energy Act of 2008?

Answer: To increase the use of local renewable energy

18. How does climate change affect Philippine farming?

Answer: Unpredictable weather damages crops and threatens food supply

19. Why are mangroves and coral reefs considered important for climate action?

Answer: They store carbon and protect coasts from storms and rising seas

20. What is the national target for renewable energy share in the Philippines by 2040?

Answer: 50%

21. Give one example each of mitigation and adaptation.

Answer: Mitigation: use solar power; Adaptation: plant flood-resistant crops

22. Why is geothermal energy especially useful for the Philippines?

Answer: It provides reliable power 24 hours a day using local resources

23. Why is adaptation still needed even if we cut emissions?

Answer: Some warming and sea-level rise are already unavoidable

24. How does expanding renewable energy help the Philippine economy?

Answer: It saves money on imported fuel and creates local jobs

25. What household action helps both reduce emissions and cope with hotter weather?

Answer: Planting shade trees or collecting rainwater

26. Is there enough renewable energy available to power the entire Philippines?

Answer: Yes; our natural resources are more than sufficient

27. What does “climate justice” mean for the Philippines?

Answer: We contribute very little to emissions but are among the most affected countries

28. How does saving electricity help reduce climate impacts?

Answer: Less coal or gas is burned, so fewer greenhouse gases are released

29. What is the main barrier to faster renewable energy adoption in the Philippines?

Answer: Existing contracts, slow approval, and grid limitations

30. Why is it important for Grade 10 students to learn about sustainability?

Answer: They will shape the future policies and face the effects of climate change